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INFORMATION PROCESSING METHOD, INFORMATION PROCESSING APPARATUS, AND PROGRAM FOR DETERMINING FUNCTION/OPERATION OF APPLIANCE

Abstract

A first-type block defined in a functional unit executable by an appliance and a second-type block defined in a unit of operation to be executed by a person are defined. A receiver receives information related to the first-type block executed by the appliance. An estimator estimates a block sequence in which the first-type block and the second-type block are arranged in an order of operation based on the received information related to the first-type block. A modifier modifies, in a case of receiving an instruction to modify the estimated block sequence, the block sequence.

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Background/Summary

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2022-190359, filed on Nov. 29, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to information processing technology, and particularly, to an information processing method, an information processing apparatus, and a program for determining a function/operation of an appliance.

2. Description of the Related Art

[0003] An operation history of an appliance is effectively utilized to provide a user with appropriate services, such as automatic control of the appliance and provision of information.

[0004] For example, if receiving the operation history of the appliance, a service provision apparatus controls the appliance such that the appliance operates according to the operation history (see, for example, JP 2004-185612 A).

[0005] A cooking recipe includes not only operations of the appliance but also human intervention. Therefore, control content also including the human intervention as in a cooking recipe cannot be created only by the operation history of the appliance. Meanwhile, it is desirable to easily create the control content also including the human intervention.

SUMMARY

[0006] The present disclosure has been made in view of such a situation, and an object of the present disclosure is to provide a technique for easily creating control content also including human intervention.

[0007] In order to solve the above problem, an information processing method according to one aspect of the present disclosure is an information processing method in which a first-type block defined in a functional unit executable by an appliance and a second-type block defined in a unit of operation to be executed by a person are defined, the information processing method including: a step of receiving information related to the first-type block executed by the appliance; a step of estimating a block sequence in which the first-type block and the second-type block are arranged in an order of operation based on the received information related to the first-type block; and a step of modifying the block sequence in a case of receiving an instruction to modify the estimated block sequence.

[0008] Another aspect of the present disclosure is an information processing apparatus. The apparatus is an information processing apparatus in which a first-type block defined in a functional unit executable by an appliance and a second-type block defined in a unit of operation to be executed by a person are defined, the information processing apparatus including: a receiver structured to receive information related to the first-type block executed by the appliance; an estimator structured to estimate a block sequence in which the first-type block and the second-type block are arranged in an order of operation based on the received information related to the first-type block; and a modifier structured to modify the block sequence in a case of receiving an instruction to modify the estimated block sequence.

[0009] Optional combinations of the aforementioned constituting elements, and implementations of the disclosure in the form of methods, apparatuses, systems, recording mediums, and computer programs may also be practiced as additional modes of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIGS. 1A to 1G show configurations of an appliance control system according to an embodiment;

[0011] FIGS. 2A to 2E show configurations of the functional blocks used in the appliance control system of FIGS. 1A to 1G;

[0012] FIGS. 3A to 3H show configurations of the functional block sequences used in the appliance control system of FIGS. 1A to 1G;

[0013] FIG. 4 shows an overview of the operation of the appliance of FIGS. 1A to 1G;

[0014] FIG. 5 shows a configuration of the appliance control system of FIGS. 1A to 1G;

[0015] FIG. 6 shows a configuration of the appliance of FIG. 5;

[0016] FIG. 7 shows a configuration of a user apparatus of FIG. 5;

[0017] FIG. 8 shows a configuration of an information processing apparatus of FIG. 5;

[0018] FIGS. 9A to 9D show cooking steps when four dishes are cooked at the same time;

[0019] FIG. 10 shows a data structure of an execution history DB of FIG. 5;

[0020] FIG. 11 shows an execution history displayed on a display of FIG. 7;

[0021] FIG. 12 shows a processing overview of a selector of FIG. 8;

[0022] FIG. 13 shows a data structure of a sample DB of FIG. 5;

[0023] FIG. 14 shows a data structure of a database stored in a storage of FIG. 5;

[0024] FIG. 15 shows a processing overview of an estimator of FIG. 8;

[0025] FIG. 16 shows a processing overview of a modifier of FIG. 8;

[0026] FIG. 17 shows another processing overview of the modifier of FIG. 8;

[0027] FIG. 18 is a flowchart showing a procedure for storing the execution history by the appliance control system of FIG. 5; and

[0028] FIG. 19 is a flowchart showing a procedure for creating a block sequence by the appliance control system of FIG. 5.

DETAILED DESCRIPTION

[0029] The invention will now be described by reference to the preferred embodiments. This does not intend to limit the scope of the present invention, but to exemplify the invention.

[0030] All of the embodiments described below show preferred embodiments of the present disclosure. Therefore, numerical values, shapes, materials, constituting elements, positions of arrangement and connection forms of constituting elements, and steps and order of steps shown in the following embodiments are examples only and are not presented to limit the present disclosure. Therefore, those of the components in the following embodiments not defined in the independent claims, which present the highest-level concept of the present disclosure, are described as optional constituting elements. Substantially identical features shown in the figures are denoted by identical symbols, and a duplicate description is omitted or simplified. Hereinafter, the embodiment will be described in the order of (1) overviews of functional blocks and functional block sequences, (2) configurations of appliance control system, (3) storage of execution history, and (4) creation of functional block sequences.

(1) Overviews of Functional Blocks and Functional Block Sequences

[0031] In household electrical/mechanical appliances such as a rice cooker, washing machine, and microwave oven (hereinafter referred to as “appliances”), the function/operation of hardware is controlled by software for realizing a specific function. In this embodiment, an appliance control

system is introduced as a mechanism to enable creation or updating of software for controlling the appliance.

[0032] FIGS. 1A to 1G show configurations of an appliance control system **1000**. In the appliance control system **1000**, a four-layer model composed of the first to fourth layers is defined. In the first layer, the configuration of an appliance **100** is defined. The appliance **100** is, for example, a rice cooker (appliance **100a**), a washing machine (appliance **100b**), and a microwave oven (appliance **100c**). The appliance **100** is not limited thereto. Each appliance **100** includes a block **2** (FIG. 1E), a block **4** (FIG. 1F), a block **6** (FIG. 1G), a plurality of components **102**, a plurality of drivers **104**, and a plurality of functional blocks **110**.

[0033] The component **102** is a hardware element constituting a unit derived from dividing the operation (actuation/sensing) of the appliance **100** and includes an actuator and a sensor that execute the function of the appliance **100**. The actuator is an output device and the sensor is an input device. The actuator includes, for example, a bottom IH (Induction Heating) coil (component **102a**), a body IH coil (component **102b**), a stepping motor (component **102c**), a water bowl IH coil (component **102d**), a cooling fan (component **102e**), and a piezoelectric buzzer (component **102f**) in the rice cooker (appliance **100a**). The sensor is, for example, a temperature sensor (component **102g**) in the rice cooker (appliance **100a**). The component **102** included in the rice cooker (appliance **100a**) is not limited to these, and the washing machine (appliance **100b**) and the microwave oven (appliance **100c**) are similarly configured.

[0034] The driver **104** is software for directly controlling the component **102**. IH control (driver **104**) in the rice cooker (appliance **100a**) controls the bottom IH coil (component **102a**). Further, IH control (driver **104b**) controls the body IH coil (component **102b**), pressure valve control (driver **104c**) controls the stepping motor (component **102c**), and IH control (driver **104d**) controls the water bowl IH coil (component **102d**). Further, fan control (driver **104e**) controls the cooling fan (component **102e**), buzzer control (driver **104f**) controls the piezoelectric buzzer (component **102f**), and sensor control (driver **104g**) controls the temperature sensor (component **102g**). The driver **104** included in the rice cooker (appliance **100a**) is not limited to these, and the washing machine (appliance **100b**) and the microwave oven (appliance **100c**) are similarly configured.

[0035] The functional block **110** is a software interface (API: Application Programming Interface) associated with one or more drivers **104** to cause one or more components **102** to operate. The functional block **110** can receive one or more parameters for controlling the operation of the (each) component **102**. Details of the functional block **110** will be described later.

[0036] In the second layer, a functional block sequence **120** in which one or more functional blocks **110** are arranged in the order of operation is defined to cause the appliance **100** to execute an intended process. That is, the functional block sequence **120** defines the order of execution of one or more functional blocks **110**. The intended process is defined according to the appliance **100**, and is, for example, cooking in the case of the rice cooker (appliance **100a**) and the microwave oven (appliance **100c**) and washing in the case of the washing machine (appliance **100b**). The functional block sequence **120a** (FIG. 1B) is used in the rice cooker (appliance **100a**), the functional block sequence **120b** (FIG. 1C) is used in the washing machine (appliance **100b**), and the functional block sequence **120c** (FIG. 1D) is used in the microwave oven (appliance **100c**). The appliance **100** executes the operation in the order of the functional blocks **110** arranged in the functional block sequence **120**. Therefore, it is possible to update the function/operation of the appliance **100** by changing the arrangement of the functional blocks **110** or changing a parameter set in the functional block **110**. Details of the functional block sequence **120** will be described later.

[0037] In the third layer, a platform server **130** that manages various information in the appliance control system **1000** is arranged. The platform server **130** includes a sequence manager, a device manager, and various databases. The sequence manager manages the functional block sequence **120**, the device manager manages the registered information on the appliance **100** that can use the functional block sequence **120**, and the various databases manage user information on users who

can use the functional block sequence **120**.

[0038] In the fourth layer, a user application server **132** in which each functional block sequence **120** is presented as a user application is arranged. The functional block sequence **120** presented in the user application server **132** is downloaded to the appliance **100**. The downloaded functional block sequence **120** is enabled in the appliance **100**. When a further functional block sequence **120** is downloaded to the appliance **100**, the further functional block sequence **120** is enabled in the appliance **100**.

[0039] The third and fourth layers of the appliance control system **1000** may be integrated. In that process, the platform server **130** and the user application server **132** are integrally configured. Alternatively, the third and fourth layers in the appliance control system **1000** may be arranged in the same layer. Alternatively, the third and fourth layers in the appliance control system **1000** may be omitted. In that process, the functional block sequence **120** is downloaded to the appliance **100** from the user apparatus (not shown) owned by the user.

[0040] FIGS. 2A to 2E show configurations of the functional block **110** used in the appliance control system **1000**. FIG. 2A shows the basic configuration of the functional block **110**. The functional block **110** is defined in a functional unit that the appliance **100** can execute and has a “block name” determined by the detail of the function. A plurality of parameters determined by the function can be set in the functional block **110**. Each parameter set in the functional block **110** is output to the driver **104**. When the driver **104** receives the parameter from the functional block **110**, the driver **104** controls the operation of the component **102** according to the parameter.

[0041] FIG. 2B shows a functional block **110a** of “pre-cook” in the rice cooker (appliance **100a**) of FIG. 1A. In the functional block **110a** of “pre-cook”, parameters including pot bottom temperature, duration, convection pattern, bottom (outside) IH time, and bottom (inside) IH time can be set. FIG. 2C shows a functional block **110b** of “boil” in the rice cooker (appliance **100a**) of FIG. 1A, FIG. 2D shows a functional block **110c** of “steam” in the rice cooker (appliance **100a**) of FIG. 1A, and FIG. 2E shows a functional block **110d** of “keep warm”. A plurality of parameters can be set in each of the functional block **110b** to the functional block **110d** as well. The same applies to the functional block **110** in the washing machine (appliance **100b**) and the microwave oven (appliance **100c**) of FIG. 1A.

[0042] FIGS. 3A to 3H show configurations of the functional block sequence **120** used in the appliance control system **1000** and, particularly, the functional block sequence **120a** used in the rice cooker (appliance **100a**) of FIG. 1A. FIG. 3A shows a sequence for “rice cooking”, FIG. 3B shows a sequence for “simmered food cooking”, and FIG. 3C shows a sequence for “roast beef (low temperature cooking)”.

[0043] In the sequence for “rice cooking” shown in FIG. 3A, three “pre-cook” functional blocks **110a** (FIG. 3D), a “cook” functional block **110n** (FIG. 3E), the “boil” functional block **110b** (FIG. 3F), the “steam” functional block **110c** (FIG. 3G), and the “keep warm” functional block **110d** (FIG. 3H) are arranged in order. In the three “pre-cook” functional blocks **110a**, mutually different parameters are set. By thus arranging the three “pre-cook” functional blocks **110a** in which mutually different parameters are set in order, a three-step pre-cooking can be executed.

[0044] In the sequence for “simmered food cooking” shown in FIG. 3B, the “pre-cook” functional block **110a**, the “cook” functional block **110n**, the “boil” functional block **110b**, and the “keep warm” functional block **110d** are arranged in order. The sequence for “roast beef (low temperature cooking)” shown in FIG. 3C includes the “keep warm” functional block **110d**. By thus changing the type, arrangement, and parameters of the functional block **110** used, it is possible to execute processes directed to different purposes of “rice cooking”, “simmered food cooking”, and “roast beef (low-temperature cooking)”. The same applies to the functional block sequence **120** in the washing machine (appliance **100b**) and the microwave oven (appliance **100c**) of FIG. 1A.

[0045] FIG. 4 shows an overview of the operation of the appliance **100** and, particularly, the rice cooker (appliance **100a**) of FIG. 1A. This shows the operation of the appliance **100a** according to

the sequence for “rice cooking” in FIG. 3A. In the water immersion step, the three “pre-cook” functional blocks **110a** with mutually different parameters set are executed in order so that the components **102** corresponding to them operate according to the parameters. As a result, the pot temperature increases in a stepwise manner over time. Following this, the “cook” functional block **110n**, the “boil” functional block **110b**, the “steam” functional block **110c**, and the “keep warm” functional block **110d** are executed in order so that the components **102** corresponding to them operate according to the parameters. That is, rice is cooked in the appliance **100a** by executing a plurality of functional blocks **110** in order.

[0046] In the description given so far, the functional block **110** is defined in a functional unit that can be executed by the appliance **100**, and the functional block sequence **120** in which one or more functional blocks **110** are arranged in the order of operation is defined. Such functional blocks **110** do not include an operation to be executed by a person. On the other hand, in an actual intended process, the operation to be executed by a person may also be required. To address this, the functional block **110** and the functional block sequence **120** described so far are expanded in the following. Therefore, the functional block **110** is also defined in a unit of operation to be executed by a person. By expressing the operation of the person as the functional block **110**, the operation of the person can be dealt with in the same way as the operation of the appliance **100**. In addition, the functional block **110** defined in the functional unit that can be executed by the appliance **100** is defined as a “first-type block”, and the functional block **110** defined in the unit of the operation to be executed by the person is defined as a “second-type block”. Hereinafter, the functional block **110** is used without distinguishing between the first-type block and the second-type block.

(2) Configuration of Appliance Control System

[0047] FIG. 5 shows the configuration of the appliance control system **1000**. The appliance control system **1000** includes the appliance **100a**, the appliance **100c**, an appliance **100d**, a user apparatus **150**, a network **300**, an information processing apparatus **400**, and a storage apparatus **450**. In addition, the storage apparatus **450** includes an execution history DB **460** and a sample DB **470**.

[0048] The appliance **100a** and the appliance **100c** are, for example, the rice cooker and the microwave oven of FIG. 1A. The appliance **100d** is, for example, an IH cooking heater. Each appliance **100** executes processes of the first layer and the second layer of FIGS. 1A and 1s connected to the network **300**. In addition to the appliance **100**, the user apparatus **150**, the information processing apparatus **400**, and the storage apparatus **450** are also connected to the network **300**. In the network **300**, any one of wired communication, wireless communication, and a combination of the wired communication and the wireless communication is executed between these apparatuses.

[0049] The user apparatus **150** is an apparatus used by a user who executes an intended process, for example, cooking, and is, for example, a computer, a smartphone, or a tablet terminal. The user apparatus **150** receives, from the user, information related to a dish to be cooked in the appliance **100**. The user apparatus **150** transmits the information related to the dish to the information processing apparatus **400** via the network **300**.

[0050] The information processing apparatus **400** is, for example, a server including a processor, a memory, and the like, or a computer such as a cloud server. In addition, the information processing apparatus **400** is also a user application server **132** for executing the process of the fourth layer of FIG. 1A. If receiving the information related to the dish from the user apparatus **150**, the information processing apparatus **400** selects the functional block sequence **120** corresponding to the dish, and transmits information on the functional block sequence **120** to the appliance **100** via the network **300**. If receiving the information on the functional block sequence **120** from the information processing apparatus **400**, the appliance **100** executes an operation according to the functional block sequence **120**.

[0051] In addition to such process, the information processing apparatus **400** creates and updates the functional block sequence **120** in cooperation with the storage apparatus **450**. At that time, the

user inputs various information using the user apparatus **150**. The information input by the user is reflected when the information processing apparatus **400** creates the functional block sequence **120**. [0052] The storage apparatus **450** is, for example, a hard disk drive (HDD), or a solid state drive (SSD), and is a storage region that can store electronic information. The execution history DB **460** stores an execution history of the functional block **110** in each appliance **100**, and the sample DB **470** stores a sample of the functional block sequence **120** for various recipes.

[0053] FIG. **6** shows a configuration of the appliance **100**. The appliance **100** includes the component **102**, a communicator **140**, a display **142**, an operator **144**, a processor **146**, and a storage **148**. The processor **146** includes the functional block **110** and the driver **104**. As described above, the appliance **100** is a household electrical/mechanical appliance such as a rice cooker, a washing machine, or a microwave oven. A plurality of components **102**, a plurality of drivers **104**, and a plurality of functional blocks **110** are provided as shown in FIGS. **1B** to **1G**, but only a respective one is shown here for the sake of clarity of the drawings.

[0054] The communicator **140** is connected to the network **300** and executes communication with the information processing apparatus **400** via the network **300**. For example, the communicator **140** receives the information on the functional block sequence **120** from the information processing apparatus **400**. The functional block sequence **120** includes the functional block **110** to be executed by the appliance **100** in the functional block sequence **120** for the intended process. In addition, the storage **148** stores the functional block **110** that can be executed in the appliance **100**. The processor **146** reads out the functional block **110** stored in the storage **148** based on the information on the functional block sequence **120** received by the communicator **140**, and causes the component **102** to execute the process according to the functional block sequence **120** via the driver **104**.

[0055] The display **142** displays information from the processor **146**. The operator **144** is an interface capable of receiving an input from a user, and is, for example, a button. In addition, the display **142** and the operator **144** may be integrated as a touch panel. The operator **144** outputs the received input to the processor **146**.

[0056] FIG. **7** shows a configuration of the user apparatus **150**. The user apparatus **150** includes a display **152**, an operator **154**, a processor **156**, a storage **158**, and a communicator **160**. As described above, the user apparatus **150** is a computer, a smartphone, or a tablet terminal. The display **152** displays information received from the processor **156**. The operator **154** is an interface capable of receiving an input from a user, and is, for example, a button. In addition, the display **152** and the operator **154** may be integrated as a touch panel. The operator **154** outputs the received input to the processor **156**.

[0057] The processor **156** outputs the information to be displayed to the display **152** and receives the input from the operator **154**. In addition, the processor **156** also stores the information in the storage **158** and reads out the information from the storage **158**. Further, the processor **156** is connected to the network **300** via the communicator **160** and executes the communication with the information processing apparatus **400** via the network **300**. With such configuration, if the operator **154** receives predetermined information from the user, the processor **156** transmits the predetermined information from the communicator **160** to the information processing apparatus **400**.

[0058] FIG. **8** shows a configuration of the information processing apparatus **400**. The information processing apparatus **400** includes a processor **406**, a storage **408**, and a communicator **410**. The processor **406** includes a receiver **420**, a selector **422**, an estimator **424**, and a modifier **426**.

[0059] The processor **406** executes the process in the information processing apparatus **400**, for example, the process of creating the functional block sequence **120**. Details of the process in the processor **406** will be described later. The storage **408** stores information used in the processor **406**. The communicator **410** is connected to the network **300** and communicates with the appliance **100**, the user apparatus **150**, and the storage apparatus **450** via the network **300**.

(3) Storage of Execution History

[0060] As described above, the functional block sequence **120** can be freely created by arranging the plurality of functional blocks **110**. However, the user does not understand the operation of each component **102** in detail. Therefore, it is difficult to understand which functional block **110** is combined to implement the operation suitable for a demand of the user. As a result, it is difficult to create the functional block sequence **120**.

[0061] For such user, it is desirable that the functional block sequence **120** be created to reproduce his/her actual cooking. For example, the functional block sequence **120** is created by storing the functional blocks **110** executed in the appliance **100** during the actual cooking and arranging the functional blocks **110** in an order of execution. However, the actual cooking includes not only the operation of the appliance **100** but also the operation of the user, but the functional block **110** associated with the operation of the user cannot be stored. Therefore, in the process as described above, it is impossible to create the functional block sequence **120** that reproduces cooking.

[0062] In addition, even if the user desires to create a recipe (functional block sequence **120**) after the cooking, it is difficult to create the functional block sequence **120** since no executed functional block **110** is stored. Further, in the cooking, not only one dish but also a plurality of dishes may be cooked at the same time. In that case, the functional blocks **110** corresponding to each of the plurality of dishes are combined, and thus, it is difficult to create the functional block sequence **120** corresponding to one dish. In the present embodiment, the functional block sequence **120** is estimated from the functional blocks **110** executed in each of the plurality of the appliances **100**, and creation of the functional block sequence **120** is supported. Such functional block sequence **120** corresponds to the recipe.

[0063] FIGS. **9A** to **9D** show cooking steps when four dishes are cooked at the same time. Here, it is assumed a case where rice, a simmered taro, and miso soup are cooked at the same time in addition to stewed diced pork. FIG. **9A** shows a stewed diced pork cooking step. In the stewed diced pork cooking step, a “cut” functional block **110aa**, a “parboil” functional block **110ab**, a “clean” functional block **110ac**, a “mix” functional block **110ad**, and a “simmer” functional block **110ae** are executed in order. Here, the “cut” functional block **110aa**, the “clean” functional block **110ac**, and the “mix” functional block **110ad** are executed by a person. The “parboil” functional block **110ab** and the “simmer” functional block **110ae** are executed by the IH cooking heater (appliance **100d**).

[0064] FIG. **9B** shows a rice cooking step. In the rice cooking process, a “clean” functional block **110ba** and a “rice cooking” functional block **110bb** are executed in order. Here, the “clean” functional block **110ba** is executed by a person, and the “rice cooking” functional block **110bb** is executed by the rice cooker (appliance **100a**).

[0065] FIG. **9C** shows a simmered taro cooking step. In the simmered taro cooking process, a “parboil” functional block **110ca**, a “peel” functional block **110cb**, a “mix” functional block **110cc**, a “simmer” functional block **110cd**, and a “cool” functional block **110ce** are executed in order. Here, the “parboil” functional block **110ca** is executed by the microwave oven (appliance **100c**), the “peel” functional block **110cb**, the “mix” functional block **110cc**, and the “cool” functional block **110ce** are executed by a person, and the “simmer” functional block **110cd** is executed by the IH cooking heater (appliance **100d**).

[0066] FIG. **9D** shows a miso soup cooking step. In the miso soup cooking step, a “cut” functional block **110da** and a “simmer” functional block **110db** are executed in order. Here, the “cut” functional block **110da** is executed by a person, and the “simmer” functional block **110db** is executed by the IH cooking heater (appliance **100d**).

[0067] After completion of the cooking, the rice cooker (appliance **100a**) reports, to the information processing apparatus **400** via the network **300**, that the “rice cooking” functional block **110bb** has been executed. The microwave oven (appliance **100c**) reports, to the information processing apparatus **400** via the network **300**, that the “parboil” functional block **110ca** has been executed.

The IH cooking heater (appliance **100d**) reports, to the information processing apparatus **400** via the network **300**, that the “cut” functional block **110aa** or the like has been executed. Each report includes user identification information for identifying a user (home).

[0068] If receiving the reports from the rice cooker (appliance **100a**), the microwave oven (appliance **100c**), and the IH cooking heater (appliance **100d**), the information processing apparatus **400** stores each report in the execution history DB **460**. FIG. **10** shows a data structure of the execution history DB **460**. In the execution history DB **460**, execution time, a name of the executed functional block **110**, a setting parameter in the functional block **110**, and the user identification information are stored for each report. Such reports are intermingled for a plurality of dishes and a plurality of appliances **100**. In addition, in a case where the execution history DB **460** stores reports from the appliances **100** used by a plurality of users, the user identification information is used to identify a transmission source of each report. In this way, the execution history DB **460** can store information related to the functional block **110** executed by the appliance **100**, but cannot store information related to the functional block **110** executed by a person.

(4) Creation of Functional Block Sequences

[0069] Here, a case is assumed where the user desires to leave the recipe of the stewed diced pork as the functional block sequence **120** after the completion of the cooking of the stewed diced pork, the rice, the simmered taro, and the miso soup as described above. Therefore, here, a process of creating the functional block sequence **120** of the stewed diced pork will be described. The user manipulates the operator **154** of the user apparatus **150** to input an instruction (hereinafter, referred to as a “creation start instruction”) to start creation of the functional block sequence **120**. The creation start instruction includes the user identification information. The communicator **160** transmits the creation start instruction to the information processing apparatus **400** via the network **300**.

[0070] The communicator **410** of the information processing apparatus **400** receives the creation start instruction from the user apparatus **150**. If the receiver **420** receives the creation start instruction, the processor **406** accesses the execution history DB **460** via the communicator **410** to acquire an execution history (one or more reports) for the user identification information included in the creation start instruction. In response to this, the receiver **420** receives the execution history from the execution history DB **460** via the communicator **410**. The execution history can be said to be the information related to the functional block **110** (first-type block) executed by the appliance **100**. The communicator **410** transmits the execution history to the user apparatus **150** via the network **300**.

[0071] The communicator **160** of the user apparatus **150** receives the execution history from the information processing apparatus **400**. The processor **156** displays the execution history on the display **152**. FIG. **11** shows an execution history displayed on the display **152**. The plurality of functional blocks **110** executed in each appliance **100** are arranged in an order of the execution time. For example, the “parboil” functional block **110ab**, the “rice cooking” functional block **110bb**, the “parboil” functional block **110ca**, and the “simmer” functional block **110cd** are arranged in order from the front. In addition, the “simmer” functional block **110ae** is arranged after 15 minutes or more from the “simmer” functional block **110cd**. Further, the “simmer” functional block **110db** is arranged after 10 minutes or more from the “simmer” functional block **110ae**.

[0072] As described above, since the user desires to leave the stewed diced pork recipe as the functional block sequence **120**, the user manipulates the operator **154** of the user apparatus **150** to select the functional block **110** included in the stewed diced pork recipe from the plurality of functional blocks **110** displayed as shown in FIG. **11**. The processor **156** receives information on the functional block **110** selected by the user. The communicator **160** transmits the information on the selected functional block **110** to the information processing apparatus **400** via the network **300**.

[0073] The communicator **410** of the information processing apparatus **400** receives the information on the selected functional block **110** from the user apparatus **150**. The receiver **420**

receives the information on the selected functional block **110**. The selector **422** selects one or more functional blocks **110** from the plurality of functional blocks **110** included in the execution history according to the information on the selected functional block **110**. This corresponds to selecting the functional block **110** corresponding to one recipe (stewed diced pork) among a plurality of recipes from the plurality of functional blocks **110** included in the execution history. FIG. **12** shows a processing overview of the selector **422**. Among the plurality of functional blocks **110** shown in FIG. **11**, the “parboil” functional block **110ab** and the “simmer” functional block **110ae** included in a stewed diced pork recipe are selected.

[0074] As described above, the user manipulates the operator **154** of the user apparatus **150** to select the functional block **110**, but may further input information related to an executed recipe. The information related to the executed recipe is, for example, at least one of a recipe name, a recipe category, and main ingredients, but is not limited thereto. The processor **156** receives the information related to the executed recipe. The communicator **160** transmits the information related to the executed recipe to the information processing apparatus **400** via the network **300**. The communicator **410** of the information processing apparatus **400** receives the information related to the executed recipe from the user apparatus **150**. The receiver **420** receives the information related to the executed recipe.

[0075] The functional block **110** selected by the selector **422** is the functional block **110** executed by the appliance **100**. Therefore, in order to reproduce the recipe by the plurality of functional blocks **110**, the functional blocks **110** executed by a person are insufficient. Based on the selected functional block **110**, the estimator **424** estimates the functional block sequence **120** in which the functional block **110** defined in the functional unit executable by the appliance **100** and the functional block **110** defined in the unit of the operation to be executed by the person are arranged in the order of operation. The estimator **424** uses the sample DB **470** for the estimation.

[0076] FIG. **13** shows a data structure of the sample DB **470**. The sample DB **470** stores a sample of the functional block sequence **120** for each of the plurality of recipes. For example, “stewed diced pork”, “rice”, “simmered taro”, “miso soup”, “A”, and the like are shown as recipe names, and the plurality of functional blocks **110** included in the sample of the functional block sequence **120** are shown respectively. Here, the sample of the functional block sequence **120** is associated with the recipe names, but may be associated with the recipe categories and the main ingredients.

[0077] The estimator **424** acquires the sample of the functional block sequence **120** corresponding to the recipe indicated in the information related to the recipe by referring to the sample DB **470** based on the information related to the recipe received by the receiver **420**. In addition, the estimator **424** specifies the functional block **110** corresponding to the functional block **110** selected by the selector **422** among the plurality of functional blocks **110** included in the acquired sample of the functional block sequence **120**. The functional block **110** corresponding to the selected functional block **110** is the same functional block **110** as the selected functional block **110** or a functional block **110** similar to the selected functional block **110**.

[0078] FIG. **14** is used herein to describe the process of identifying the functional block **110** similar to the selected functional block **110**. FIG. **14** shows a data structure of a database stored in the storage **408**. The functional block **110** of “B1”, the functional block **110** of “B2”, . . . , and the functional block **110** of “BN” having similar functions are grouped as a “group B”. A plurality of types of such groups are defined. It can be said that other functional blocks **110** in the group including the selected functional block **110** are functional blocks **110** similar to the selected functional block **110**.

[0079] FIG. **15** shows a processing overview of the estimator **424**. This indicates the functional block sequence **120** estimated by the estimator **424**. A “cut” functional block **110ea**, a “parboil” functional block **110ab**, a “clean” functional block **110ec**, a “mix” functional block **110ed**, and a “simmer” functional block **110ae** are arranged in order from the front. Here, the “parboil” functional block **110ab** and the “simmer” functional block **110ae** are the functional blocks **110**

included in the execution history, and the “cut” functional block **110ea**, the “clean” functional block **110ec**, and the “mix” functional block **110ed** are the functional blocks **110** estimated by the estimator **424**. By such process, the functional block sequence **120** also including the functional block **110** associated with the operation of the user is estimated from the functional block **110** associated with the operation of the appliance **100**.

[0080] In a case where the user does not input the information related to the executed recipe to the user apparatus **150**, the receiver **420** of the information processing apparatus **400** does not receive the information related to the executed recipe. In a case where the receiver **420** does not receive the information related to the executed recipe, the estimator **424** acquires the sample of the functional block sequence **120** including a pattern by referring to the sample DB **470** based on the pattern in which the functional blocks **110** selected by the selector **422** are arranged. That is, the estimator **424** acquires the sample of the functional block sequence **120** including the same pattern as the pattern in which the functional blocks **110** selected by the selector **422** are arranged. At that time, the estimator **424** may use information on a time interval between the selected functional blocks **110**. In addition, information on the ingredients, the appliance **100**, and information on the setting parameters (strength/time) may be used. A subsequent process may be the same as before, and the estimator **424** estimates the functional block sequence **120** as shown in FIG. **15**. The communicator **410** transmits the information on the functional block sequence **120** estimated by the estimator **424** to the user apparatus **150** via the network **300**.

[0081] The communicator **160** of the user apparatus **150** receives the information on the functional block sequence **120** from the information processing apparatus **400**. The processor **156** displays the information on the functional block sequence **120** as shown in FIG. **15** on the display **152**. The user manipulates the operator **154** to input a modification instruction for the functional block sequence **120** while looking at the functional block sequence **120** displayed on the display **152**. The modification instruction is, for example, deletion or change of the functional block **110** included in the functional block sequence **120**, addition of the functional block **110** to the functional block sequence **120**, or input of a setting parameter (for example, an ingredient or a gram).

[0082] The processor **156** receives the modification instruction. The communicator **160** transmits the modification instruction to the information processing apparatus **400** via the network **300**. The communicator **410** of the information processing apparatus **400** receives the modification instruction from the user apparatus **150**. The receiver **420** receives the modification instruction. In a case of receiving the modification instruction, the modifier **426** modifies the functional block sequence **120** according to the modification instruction. FIG. **16** shows a processing overview of the modifier **426**. Here, as an example, a setting parameter is added to the functional block **110ea** and the like in the functional block sequence **120** shown in FIG. **15**. The modification in the modifier **426** is not limited thereto.

[0083] The modifier **426** may specify a functional block **110** similar to the functional block **110** included in the functional block sequence **120**, that is, a functional block **110** having an equivalent function with reference to the table of FIG. **14**, and propose conversion into such functional block **110**. FIG. **17** shows another processing overview of the modifier **426**. Here, a “simmer” functional block **110ef** having an equivalent function to the “simmer” functional block **110ae** is a candidate for the conversion. The conversion is proposed on the display **152** of the user apparatus **150**, and in a case of the conversion, the user inputs a conversion instruction to the operator **154** of the user apparatus **150**. If the receiver **420** of the information processing apparatus **400** receives the conversion instruction, the modifier **426** converts the functional block **110**. In a case where the functional block sequence **120** is modified, the storage **408** stores both the functional block sequence **120** before the modification and the functional block sequence **120** after the modification.

[0084] The features are implemented in hardware such as a Central Processing Unit (CPU), a memory, or other Large Scale Integrations (LSIs) of any computer and in software such as a program loaded into a memory. The figure depicts functional blocks implemented by the

cooperation of these elements. Therefore, it will be understood by those skilled in the art that these functional blocks may be implemented in a variety of manners by hardware only or by a combination of hardware and software.

[0085] An operation of the appliance control system **1000** having the above configuration will be described. FIG. **18** is a flowchart showing a procedure for storing an execution history by the appliance control system **1000**. The user sets the appliance **100** (S10). The user instructs the appliance **100** to start using the appliance **100** (S12). The functional block **110** is executed in the appliance **100** (S14). The appliance **100** adds the execution history of the functional block **110** to the execution history DB **460** (S16).

[0086] FIG. **19** is a flowchart showing a procedure for creating a block sequence by the appliance control system **1000**. The creation start instruction of the functional block sequence **120** by the user is input to the information processing apparatus **400** via the user apparatus **150** (S50). The information processing apparatus **400** causes the user apparatus **150** to display a most recently executed functional block **110** (S52). The information is input by the user to the information processing apparatus **400** via the user apparatus **150** (S54). The information processing apparatus **400** causes the user apparatus **150** to display the functional blocks **110** narrowed down by the user (S56). The information processing apparatus **400** estimates the functional block sequence **120** based on the narrowed-down functional block **110** (S58), and causes the user apparatus **150** to display the functional block sequence **120** (S60). The information processing apparatus **400** receives the modification by the user via the user apparatus **150** (S62), and modifies the functional block sequence **120** (S64).

[0087] According to the present embodiment, since the functional block sequence **120** in which the first-type block and the second-type block are arranged in the order of operation is estimated based on the received information related to the first-type block, it is possible to easily create the control content also including the human intervention. In addition, in a case where the modification instruction for the estimated functional block sequence **120** is received, the functional block sequence **120** is modified, and thus, it is possible to easily create the functional block sequence **120** that meets the user's desire. In addition, it is not necessary to plan the cooking systematically for creating the functional block sequence **120**, and the functional block sequence **120** can be easily created only by selecting the functional blocks **110** from one's own cooking history.

[0088] In addition, if the information related to the functional block **110** corresponding to one recipe among the plurality of recipes is selected from the execution history, the functional block sequence **120** is estimated based on the selected functional block **110**. Therefore, even if the plurality of recipes is intermingled in the execution history, the functional block sequence **120** for a desired recipe can be created. Further, the sample of the functional block sequence **120** is acquired by referring to the sample DB **470** based on the pattern in which the functional blocks **110** are arranged, and the functional block sequence **120** is estimated using the sample, so that the functional blocks **110** related to the operation of the person can be added. In addition, the sample of the functional block sequence **120** is acquired by referring to the sample DB **470** based on the received information related to the recipe, and the functional block sequence **120** is estimated using the sample, so that the functional blocks **110** related to the operation of the person can be added.

[0089] An overview of one aspect of the present disclosure is as follows. An information processing method according to one aspect of the present disclosure is an information processing method in which a first-type block defined in a functional unit executable by an appliance (**100**) and a second-type block defined in a unit of operation to be executed by a person are defined, the information processing method including: a step of receiving information related to the first-type block executed by the appliance (**100**); a step of estimating a block sequence in which the first-type block and the second-type block are arranged in an order of operation based on the received information related to the first-type block; and a step of modifying the block sequence in a case of receiving an instruction to modify the estimated block sequence.

[0090] The method further includes a step of selecting information related to the first-type block corresponding to one recipe among a plurality of recipes from the received information related to the first-type block. In the estimation step, the block sequence may be estimated based on the selected information related to the first-type block.

[0091] In the estimation step, a sample of the block sequence may be acquired by referring to a storage region in which the sample of the block sequence for each of the plurality of recipes is stored based on a pattern in which the first-type block is arranged, and the block sequence may be estimated based on the acquired sample of the block sequence and the information related to the first-type block.

[0092] The method further includes a step of receiving information related to an executed recipe. In the estimation step, the sample of the block sequence may be acquired by referring to a storage region (470) in which the sample of the block sequence for each of the plurality of recipes is stored based on the received information related to the recipe, and the block sequence may be estimated based on the acquired sample of the block sequence and the information related to the first-type block.

[0093] Another aspect of the present disclosure is an information processing apparatus (400). The apparatus is an information processing apparatus (400) in which a first-type block defined in a functional unit executable by an appliance (100) and a second-type block defined in a unit of operation to be executed by a person are defined, the information processing apparatus (400) including: a receiver (420) structured to receive information related to the first-type block executed by the appliance (100); an estimator (424) structured to estimate a block sequence in which the first-type block and the second-type block are arranged in an order of operation based on the received information related to the first-type block; and a modifier (426) structured to modify the block sequence in a case of receiving an instruction to modify the estimated block sequence.

[0094] The present disclosure has been described above based on the embodiments. It is to be understood by a person skilled in the art that the embodiments are examples, various modifications can be made to combinations of the respective components or the respective processing processes, and such modifications are also within the scope of the present disclosure.

Claims

1. An information processing method in which a first-type block defined in a functional unit executable by an appliance and a second-type block defined in a unit of operation to be executed by a person are defined, the information processing method comprising: receiving information related to the first-type block executed by the appliance; estimating a block sequence in which the first-type block and the second-type block are arranged in an order of operation based on the received information related to the first-type block; and modifying the block sequence in a case of receiving an instruction to modify the estimated block sequence.

2. The information processing method according to claim 1, further comprising selecting information related to the first-type block corresponding to one recipe among a plurality of recipes from the received information related to the first-type block, wherein in the estimating, the block sequence is estimated based on the selected information related to the first-type block.

3. The information processing method according to claim 1, wherein in the estimating, a sample of the block sequence is acquired by referring to a storage region in which the sample of the block sequence for each of the plurality of recipes is stored based on a pattern in which the first-type block is arranged, and the block sequence is estimated based on the acquired sample of the block sequence and the information related to the first-type block.

4. The information processing method according to claim 1, further comprising receiving information related to an executed recipe, wherein in the estimating, a sample of the block sequence is acquired by referring to a storage region in which the sample of the block sequence for

each of the plurality of recipes is stored based on the received information related to the recipe, and the block sequence is estimated based on the acquired sample of the block sequence and the information related to the first-type block.

5. An information processing apparatus in which a first-type block defined in a functional unit executable by an appliance and a second-type block defined in a unit of operation to be executed by a person are defined, the information processing apparatus comprising: a receiver structured to receive information related to the first-type block executed by the appliance; an estimator structured to estimate a block sequence in which the first-type block and the second-type block are arranged in an order of operation based on the received information related to the first-type block; and a modifier structured to modify the block sequence in a case of receiving an instruction to modify the estimated block sequence.

6. A program in which a first-type block defined in a functional unit executable by an appliance and a second-type block defined in a unit of operation to be executed by a person are defined, the program causing a computer to: receive information related to the first-type block executed by the appliance; estimate a block sequence in which the first-type block and the second-type block are arranged in an order of operation based on the received information related to the first-type block; and modify the block sequence in a case of receiving an instruction to modify the estimated block sequence.
